Sina **Amirrajab**

ARTIFICIAL INTELLIGENCE RESEARCHER

Eindhoven, The Netherlands

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"Highly organized, passionate, self-motivated, and multidisciplinary researcher with extensive proven experience in Artificial Intelligence and Generative AI for radiology and medical image analysis."

My interactive online CV: **sinaamirrajab.github.io**

Skills

Expertise	AI and Generative AI for Radiology and Medical Imaging; Cardiovascular MR; Cancer CT Imaging; Foundation Models; LLMs; RAG; Multi-modal Vision-Language Models; Multi-modal Fusion; Latent Diffusion Models; Healthcare applications including disease diagnosis, image segmentation, anomaly detection, prognosis, survival analysis, and treatment selection
AI/ML Tools	Python, PyTorch, MONAI, PyTorch Lightning, Gradio, scikit-learn, scikit-image, OpenCV, Weights & Biases (WandB), NumPy, Pandas, Matplotlib, Plotly, Seaborn, TIMM, Ollama, Hugging Face Transformers
Languages	English, Persian, Dutch

Education

Ph.D. in Biomedical Engineering (Marie Curie Fellow)
Eindhoven, The Netherlands

EINDHOVEN UNIVERSITY OF TECHNOLOGY — *2018 - 2022*

- Focus: AI for Cardiac MRI — **OpenGTN Link** (Marie Curie ITN-EID)
- Thesis: Cardiac MR Image Simulation and Synthesis for Medical Image Analysis (link)
- Supervisors: Prof. M. Breeuwer, Prof. J. Pluim, Dr. J. Weese, Dr. C. Lorenz

M.Sc. in Biomedical Engineering
Tehran, Iran

AMIRKABIR UNIVERSITY OF TECHNOLOGY — *2014 - 2017*

- Focus: Metal artifact estimation and correction in MRI
- Supervisor: Prof. A. Nasiraei Moghaddam

B.Sc. in Electrical Engineering (Electronics)
Rasht, Iran

UNIVERSITY OF GUILAN — *2009 - 2013*

- Focus: Solid State Relay design and production
- Supervisor: Prof. S. M. Hosseini-Golgoo

Professional Experience

Postdoctoral Researcher
Maastricht, The Netherlands

MAASTRICHT UNIVERSITY — *2024 – present*

- Research focus on AI, Generative AI, Foundation Models, and multi-modal Vision–Language Models for medical imaging.
- Lead grant writing, project design, and PhD supervision in areas including conditional image synthesis, large-scale foundation models, and multi-modal vision–language integration.
- Team: Prof. Dr. P. Lambin, The D-Lab Dpt of Precision Medicine, FHML.

Clinical AI Scientist
Münster, Germany

UNIVERSITY HOSPITAL MÜNSTER — *2024 – present*

- Deploying open-source, privacy-preserving LLMs for automated clinical data curation and workflow optimization.
- Developing diagnostic AI models to enable robust, accurate, and automated cardiovascular analysis from cardiac MRI.

Postdoctoral Researcher
Eindhoven, The Netherlands

EINDHOVEN UNIVERSITY OF TECHNOLOGY — *2022 - 2024*

- Focus: AI for Magnetic Resonance Spectroscopy — **Spectralligence Link** (ITEA4-funded)
- Project leadership and PhD/master's supervision on deep learning for medical imaging, generative models, and image super-resolution

AI Researcher in Medical Image Analysis
Eindhoven, The Netherlands

EINDHOVEN UNIVERISTY OF TECHNOLOGY — *2018 - 2024*

- Topics: Generative AI for medical image analysis, deep learning, Artificial Intelligence, Image segmentation, Image simulation and synthesis, Cardiac Magnetic Resonance Imaging, MRI acquisition and Reconstruction, MR Spectroscopy analysis, and Anomaly detection.

Undergraduate Research Advisor

Amirkabir University of Technology

TEHRAN, IRAN — *2017 - 2018*

- Design and Production of a Smart Contrast Agent Injection Pump used with MR Angiography.

MR Researcher

Amiran Negar Teb Company, Tehran, Iran.

TEHRAN, IRAN — *2016 - 2018*

- Research and development in MR physics.
- Establishment of quality assurance procedure for MRI based on the ACR Accreditation Program.
- Performing quality control and acceptance testing for newly purchased MRI scanners for institutes and hospitals.

Research Assistant

Advanced Medical Imaging Research Lab (AMIRLab)

TEHRAN, IRAN — *2014 - 2017*

- Simulating magnetic field perturbation in presence of metal with ANSYS Maxwell and MRI procedure in presence of static magnetic field inhomogeneity.
- Pulse sequence simulation with JEMRIS and MRiLab.
- Specific phantom design to perform MRI and modify the parameter to reach less metal-related artifact.

Electronics Research and Development Engineer

Persian Electronics Company

TEHRAN, IRAN — *2013 - 2015*

- Industrial project advisor and planning designer.
- Electrical PCB designer for advanced control systems.
- Product management and quality control.

Teaching and Supervision

UM

- PhD co-supervisor: Yumeng Zhang, Yi Chen
- MSc co-supervisor: Cédric Cortenraede
- MSc course teaching: Imaging Informatics

TUe

- PhD co-supervisor: Evi MC Huijben - Medical Image Synthesis with Deep Learning for Clinical Applications (**Link**)
- PhD co-supervisor: Dennis MJ van de Sande - Spectralligence AI211009 (**Link**)
- PhD co-supervisor: Stephanie Gonzalez Riedel - Educational Magnetic Resonance Imaging Simulator (eduMRIsim) (**Link**)
- MSc co-supervisor: Evianne Kruithof, Didier Lustermans, Rien Boonstoppel
- MSc course teaching: Capita Selecta in Medical Image Analysis

Professional Certificates and Awards

2018	**(Link) Marie Sklodowska-Curie Actions (MSCA) Horizon 2020 fellowship**,
2020	**(Link) Interview about my research in generative models for synthesizing realistic cardiac MRI**,
2021	**(Link) GANs Specialization Coursera**,
2021	**(Link) Stress Perfusion Imaging King's College London**,
2021	**(Link) Philips Best Paper Award on Scientific Impact Computer Vision**,
2022	**(Link) CMRxMotion Challenge STACOM Workshop 3rd Place Award: Image Quality Assessment** ,
2022	**(Link) CMRxMotion Challenge STACOM Workshop 3rd Place Award: Robust CMR Segmentation**,
2023	**(Link) Edited Magnetic Resonance Spectroscopy Challenge 2nd Place**,
2025	**(Link) 1st place in the VLM3D Challenge of MICCAI for generating high-resolution synthetic CT images from radiological reports**,
2025	**(Link) 3rd place in the Method Track of the FOMO25 - Foundation Model Challenge for Brain MRI**,

Publications

A complete list of publications can be found in my Google Scholar: LINK